

RETAIL BASKET MODEL

Energy-based basket intelligence for retail

One probability model for recommendation, generation and decision support

Version 4 model and current empirical interpretation

The model turns transactions into basket intelligence

Understand

Learn what different customers buy, which products appear together, and how basket composition changes across contexts.

Predict

Complete a basket, rank candidate products, and generate plausible baskets for a known shopping context.

Plan

Screen bundles, prices and promotions before spending money on controlled retailer experiments.

Current position: predictive basket intelligence, not autonomous pricing.

The retailer receives five practical capabilities

1. **Basket recommendation**

Rank products that best complete a customer's current basket.

2. **Bundle discovery**

Find supported product relationships beyond popularity alone.

3. **Customer understanding**

Summarize preferences, basket patterns and apparent price sensitivity.

4. **Scenario screening**

Compare predicted basket response under candidate prices or promotions.

5. **Basket simulation**

Generate baskets for testing, planning and model diagnostics.

Business use: reduce thousands of possible actions to a small, evidence-based shortlist.

The problem is to model a complete basket, not isolated products

For a known shopping context x , we want one normalized probability for every feasible non-empty basket S :

$$p(S \mid x, \text{purchase trip})$$

Inputs

Customer, prices, promotions, calendar, store, available catalogue and declared history.

Current data conditions on a purchase trip; predicting a visit requires a separate model.

Output

A probability distribution over sets of distinct products, with basket sizes from 1 to 120.

Every basket receives an energy, then energies become probabilities

$$p_{\Theta}(S \mid x) = \frac{\exp\{E_{\Theta}(S; x)\}}{Z_{+}(x)}, \quad Z_{+}(x) = \sum_{T \in \Omega_x^{+}} \exp\{E_{\Theta}(T; x)\}.$$

- S : the basket being scored
- x : the observed shopping context
- $E_{\Theta}(S; x)$: basket energy or compatibility score
- Θ : all learned model parameters
- $\mathcal{A}_x$: products offered in context x
- Ω_x^{+} : all non-empty subsets of $\mathcal{A}_x$ up to size 120
- $Z_{+}(x)$: the normalizer over every feasible basket
- Higher energy means higher probability after normalization

The energy combines product value, interaction and basket shape

$$E_{\Theta}(S; x) = \sum_{j \in S} b_j(x) + \sum_{j < k; j, k \in S} \phi_j^{\top} \phi_k - \sum_c \rho_c \binom{n_c(S)}{2} - \rho_0(|S|)$$

$b_j(x)$	$\phi_j^{\top} \phi_k$	$\rho_c, n_c(S)$	$\rho_0(S)$
Contextual value of product j .	Low-rank relationship between products j and k .	Category effect and number of basket products in category c .	Flexible preference over total basket size.

j, k index products, c indexes affinity groups, and $|S|$ is the number of distinct products in the basket.

Contextual utility explains why the same product differs by trip

$$b_{jht} = \lambda_j + \theta_h^\top \alpha_j - g_{hj} \Delta \log p_{jt} + w_j^{\text{dsp}} D_{jt} + w_j^{\text{mlr}} M_{jt} \\ + \mu_j^\top \delta_{w(t)} + \zeta_j^\top \xi_{s(t)} + \psi_j^\top r_{jht}.$$

- j, h, t : product, household and shopping time
- λ_j : product's baseline appeal
- $\theta_h^\top \alpha_j$: household-product preference
- $g_{hj} \geq 0$: household-product price sensitivity
- $\Delta \log p_{jt}$: change in log price
- D_{jt}, M_{jt} : display and mailer promotion indicators
- $w_j^{\text{dsp}}, w_j^{\text{mlr}}$: promotion responses
- $\mu_j^\top \delta_{w(t)}$: calendar effect
- $\zeta_j^\top \xi_{s(t)}$: store effect
- $\psi_j^\top r_{jht}$: declared history features

The only fundamental computational obstacle is the normalizer

The observed basket is easy to score. The difficulty is the denominator:

$$Z_+(x) = \sum_{T \in \Omega_x^+} \exp\{E_\Theta(T; x)\}.$$

Why direct enumeration fails

With 5,455 products, the number of possible subsets is on the scale of 2^{5455} . Training, incidence probabilities and calibrated counterfactuals all require Z_+ or its derivatives.

The tractability theorem removes exponential catalogue growth

Let $z \sim \mathcal{N}(0, I_r)$, where r is the interaction rank. Define

$$w_j(z; x) = \exp\left\{b_j(x) - \frac{1}{2}\|\phi_j\|^2 + z^\top \phi_j\right\}.$$

The Hubbard-Stratonovich and elementary-polynomial result is

$$Z_+(x) = \mathbb{E}_z \left[\sum_{n=1}^{n_{\max}} e^{-\rho_0(n)} A_n(z, x) \right].$$

$A_n(z, x)$ is the exact total conditional weight of all size- n baskets, computed by elementary symmetric polynomials and category convolution.

Result: an exact dynamic program plus an r -dimensional Gaussian integral.

Three simple identities make the theorem work

1. Rewrite all pair interactions as one squared low-rank sum

$$\sum_{j < k \in S} \phi_j^\top \phi_k = \frac{1}{2} \left\| \sum_{j \in S} \phi_j \right\|^2 - \frac{1}{2} \sum_{j \in S} \|\phi_j\|^2.$$

2. Use a Gaussian expectation for the squared term

$$\mathbb{E}_z[e^{z^\top v}] = e^{\|v\|^2/2}.$$

3. Use elementary symmetric polynomials to sum distinct products

The coefficient of degree n collects every size- n basket exactly once, including its category and total-size terms.

Training and every application use the same joint law

For an observed basket S_x , training maximizes

$$\ell_x(\Theta) = E_{\Theta}(S_x; x) - \log Z_+(x).$$

Observed signal

Raise the energy of baskets that actually occurred.

Model correction

Lower probability assigned to alternatives through $\log Z_+$.

One objective

Recommendation is not trained separately; it is a conditional query of the fitted law.

Recommendation is exact because the common normalizer cancels

Let R be the revealed basket and let product $j \notin R$ be a candidate completion. Define its add-one score

$$s_j(R, x) = E(R \cup \{j\}; x) - E(R; x).$$

Then

$$P(j \text{ completes } R \mid x, \text{one item missing}) = \frac{e^{s_j(R, x)}}{\sum_{k \notin R} e^{s_k(R, x)}}.$$

The score combines standalone utility, compatibility with products already in R , category crowding and the basket-size increment. Ranking needs no partition-function approximation.

Basket generation follows the probability law in four stages

1. Draw the latent interaction state z

This summarizes the low-rank basket interaction direction.

2. Draw total size n

Use weights $e^{-\rho_0(n)} A_n(z, x)$ for $n = 1, \dots, 120$.

3. Draw category counts

Allocate the selected total across affinity groups.

4. Draw distinct products

Sample the required number within each group using the conditional weights $w_j(z; x)$.

Current output: distinct-product baskets. Units, margin and inventory need a separate quantity model.

Incidence and price scenarios are derivatives or reweighted expectations

Product incidence

$$\pi_j(x) = P(j \in S \mid x) = \frac{\partial \log Z_+(x)}{\partial b_j}.$$

This includes utility, interactions, basket size and competition for basket positions.

Price or promotion scenario

If action a changes basket energy by $\Delta_a(S) = E_a(S) - E_f(S)$, then

$$\mathbb{E}_a[f(S)] = \frac{\mathbb{E}_f[f(S)e^{\Delta_a(S)}]}{\mathbb{E}_f[e^{\Delta_a(S)}]}.$$

Predictive response; causal only with credible identified price variation.

Interactions and segments turn probabilities into retailer insight

The identifiable interaction object is the Gram matrix $K = \Phi\Phi^\top$, where $K_{jk} = \phi_j^\top \phi_k$.

Product relationships

K_{jk} measures how product j changes product k 's add-one energy, holding the rest of the basket fixed. Category and size effects must also be included.

Customer segments

Group household preference surfaces and summarize their products, basket patterns and apparent price response. Segments describe behaviour; they are not permanent identities.

Predictive bundle evidence, not proof of causal complementarity.

Current evidence supports basket intelligence, with a generation warning

What is established

- Gain over exact additive parent: 0.03275 ± 0.00239 nats per basket.
- Gains over Bernoulli, DPP and NDPP: 2.252, 2.262 and 1.812 nats.
- Locked test MRR: 0.09525 ± 0.00607 .
- Strong interaction-pair lift: 1.216; matched controls: 0.998.

What is not yet calibrated

- Generated mean size: 7.34 versus observed 10.03.
- Generated size variance: 63.83 versus observed 136.28.
- Interaction-only recommendation gain is not yet conclusive.
- Price responses are predictive, not causal revenue estimates.

Source: locked Version 4 household-size rank-one evaluation.

Retail deployment should move from prediction to controlled learning

1. Understand

Train on historical baskets. Deliver segments, recommendations, supported product pairs and scenario rankings.

2. Test

Randomize a small set of model-supported bundles or promotions. Record exposure, non-purchase, quantity and margin.

3. Optimize

Calibrate incremental profit, add inventory and sequential state, then optimize a budget-constrained policy.

RL can optimize a calibrated sequential environment; it cannot repair a biased one.

The immediate value is a smaller, better decision space

The model gives the retailer one coherent view of basket behaviour.

It connects customer context, product value, interactions, basket size and price response in a normalized probability law.

Use it now to recommend, discover and shortlist.

Use controlled experiments to turn those predictions into decisions.
